

COOKING OIL QUALITY EVALUATION VIA DIELECTRIC CONSTANT MEASUREMENT USING CAPACITIVE SENSING

*Mini-Project Report submitted to the SASTRA Deemed to be University
in partial fulfillment of the requirements
for the award of the degree of*

B. Tech. Electronics and Communication Engineering

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This is to certify that the report titled “**Cooking Oil Quality Evaluation via Dielectric Constant Measurement Using Capacitive Sensing**” submitted in partial fulfillment of the requirements for the award of the degree of B. Tech. Electronics and Communication Engineering to the SASTRA Deemed to be University, is a bona-fide record of the work done by **Ms. Nithisri T J**(Reg. No.:126004179) ,**Ms. Ponni T**(Reg. No.:126004193) and **Ms. Shanmuga Priya R**(Reg. No.:126004246) during the Sixth semester of the academic year 2024-25, in the **School of Electrical & Electronics Engineering**, under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

Signature of Project Supervisor :

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We declare that the report titled “Cooking Oil Quality Evaluation via Dielectric Constant Measurement Using Capacitive Sensing” submitted by us is an original work done by us under the guidance of **Dr Susan D, Associate Professor, School of Electrical & Electronics Engineering, SASTRA Deemed to be University** during the Sixth semester of the academic year **2024-25**, in the **School of Electrical & Electronics Engineering**. The work is original and wherever we have used materials from other sources, we have given due credit and cited them in the text of the report. This report has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University.

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ABSTRACT

The prolonged reuse of cooking oil leads to chemical alterations that degrade its quality and pose significant health risks. Traditional oil quality assessment methods, while accurate, are often expensive and require sophisticated laboratory equipment. In this project, a cost-effective, portable, and real-time solution for evaluating cooking oil quality based on dielectric constant measurement using capacitive sensing techniques is proposed. Two primary methodologies are explored: capacitance-to-voltage and capacitance-to-frequency conversion circuits. After identifying the limitations in sensitivity and noise immunity in voltage conversion methods, a capacitance-to-frequency converter using an astable multivibrator and a custom-designed interleaved capacitive sensor is used. Calibration and validation are performed using an LCR meter to ensure measurement accuracy. Further enhancements by using a Wien bridge oscillator provided stable and sensitive detection, especially for minor capacitance variations in the Pico Farad range. Experimental results demonstrated a strong correlation ($R^2 \approx 0.9$) between the circuit output and LCR meter measurements, validating the system's reliability for differentiating between good and degraded oils. This project offers a promising approach for accessible, on-site cooking oil quality monitoring.

Specific Contribution:

Specific Learning:

Signature of guide:

Student reg no:

Name:

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CHAPTER 1

INTRODUCTION

In restaurants and commercial food facilities, oil quality is an important factor in food safety and adherence to health requirements while food is in storage or being prepared. Because the oil is reused over and over, the physical and chemical characteristics of oil can change, and it can form harmful compounds. Oil quality maintenance is thus important in preventing health risks and product compromise.

Here, a method is presented to determine the quality of oil based on its dielectric constant. The dielectric constant, a physical property of a material to hold electric energy in an electric field, is a function of the composition and purity of the oil. Oil deteriorates with usage and therefore its dielectric properties vary, and hence the dielectric constant can be used as a good parameter to find the quality of oil in a non-destructive real-time manner.

Our intention is to develop a low-cost, reliable, and simple-to-use system that will be capable of detecting oil degradation by measuring variations in the dielectric constant. This method aids everyday consumers as well as the food trade in determining how many times one can reuse oil safely, minimizing waste without compromising on health and safety.

This project intends utilization of appropriate sensors to determine the dielectric constant and examine the data in a way that draws conclusions about the quality of the oil. The outcomes of this project can enhance the methods of utilizing oil and bring awareness about food safety.

1.1 TESTO 270

Fast and accurate assessment of cooking oil quality is possible through the portable Testo 270. It measures the quantity of Total Polar Materials (TPM) in oil, an increasing amount with reuse and disintegration of oils. The device works through a sensor, capacitive in nature, that measures dielectric change in the oil, directly related to increasing accumulation of polar compounds, such as free fatty acids and oxidation products, in the used oil. These components indicate that oil is neither safe nor healthy for cooking purposes anymore. The equipment is widely popular in places such as food production companies and commercial kitchens where instant results can be obtained without complicated laboratory equipment. All that the operator has to do is, insert a probe into the oil sample one wishes to test, and this simple procedure yields a digital readout that lets one know whether the oil is still within the safe usage range.

1.2 LCR METER

An LCR meter is an electronic measuring instrument that measures inductance (L), capacitance (C), and resistance (R) of electronic components. In this project, the LCR meter is instrumental in accurately measuring oil sample capacitance, which is necessary for determining their dielectric constant. With repeated use, the quality of oil alters, and hence its dielectric properties differ. The LCR meter gives accurate measurements of such changes, enabling a proper measurement of oil degradation. In contrast to specially designed circuits, the LCR meter gives high accuracy and sensitivity, making it an ideal reference device to confirm the results obtained from the capacitive sensor circuit. By comparing both the measured dielectric constant values using the LCR meter and the circuit developed, the correctness of the circuit is verified. Accordingly, the LCR meter act as a standard measuring tool to validate the performance and calibration.

1.3 MOTIVATION

Highly cautioned quality assessment of cooking oil can be found by inspection and analysis of these oils in most residential, commercial kitchens, and food processing plants. Due to repeated use, several changes occur in the chemical structure of the cooking oil owing to its oxidation

and thermal decomposition. These changes typically yield a plethora of hazardous products such as trans fats, free fatty acids, and aldehydes. These products impart undesirable taste and aroma to food and eventually cause harmful health effects characterized by very high LDL cholesterol biosynthesis, cardiovascular diseases, and prolonged exposure to cancer. Even though advanced analytical methods for ascertaining oil quality are available, they tend to be costly, need experts, and are not practical for real-time analysis in common kitchen settings.

This project is inspired by the necessity for a low-cost, portable, and easy-to-use solution that can allow individuals and small enterprises to assess oil quality in real-time. By targeting the dielectric constant, a physical parameter that indicates the material's capacity for electrical energy storage, we would seek to build a system offering a consistent indicator of oil degradation. The dielectric constant variations follow in a predictable manner with the chemical composition of the oil, making it a strong non-destructive means of quality control. By capacitive sensing and optimized circuitry, it is possible to foresee a cost-effective and easy-to-use solution that enables people to make knowledgeable choices regarding the reuse of oil, minimizing health risks while fostering sustainability and food safety.

CHAPTER 2

LITERATURE REVIEW AND OBJECTIVES

2.1 LITERATURE REVIEW

Several research works have investigated dielectric property measurement as a tool for evaluating edible oil quality:

1. Mohd Mustafa Khan, Anwar Sadat, and Ekram Khan (2023) assessed the adulteration of olive oil based on dielectric spectroscopy and reported that the incorporation of adulterants greatly changes dielectric properties. This project generated a low-cost, real-time system for cooking oil quality evaluation using dielectric constant measurements. The system is validated to distinguish good and bad oils by correlating the circuit output with LCR meter readings and analysis of temperature effects, thus giving it an everyday practical capability for oils quality assessment.
2. The exciting work done by Hari Krishna Salila Vijayalal Mohan and Andrew Alexander Malcolm (2022) on capacitance-sensing-based technology to assess edible oil purity focused on real-time non-intrusive quality evaluations. Demonstrated here is a fast, low-cost, and non-invasive capacitive sensing system for detecting adulteration of extra virgin olive oil. With high sensitivity and fast response, it promises a practical real-time inline assessment of oil quality, to ensure food safety in its industrial applications.
3. Alfadhli Yahya Khaled et al. (2015) presented capacitive sensing to understand frying oil as it is heated, showing significant correlations between changes in capacitance and oil degradation. This project developed a capacitive sensing system to assess frying oil degradation by measuring the change in capacitance. The high level of correlation with total polar compounds and viscosity indicates that this approach is valid. The system provides a straightforward, inexpensive, and reliable way to evaluate frying oil quality in real-time.
4. Ahmed Fendri et al. (2019), Lossy Capacitive Sensor Measurement System, particularly for the evaluation of quality of edible oils, which offers accurate results even with dielectric losses by adopting a lossy capacitive sensor measurement system. This research by Ahmed Fendri et al. (2019) successfully created a portable, cost-effective capacitive sensor assessment system for quality evaluation of edible oils. The dielectric parameter such as relative permittivity and

conductivity has been shown to be effective for oil degradation monitoring and was proved reliable for oil quality monitoring during frying.

2.2 OBJECTIVES

1. Develop a cost-effective and reliable system to assess cooking oil quality using dielectric constant measurements by leveraging capacitive sensing and affordable electronic components suitable for real-world use.
2. Design and test circuits capable of converting small capacitance changes into measurable voltage and frequency outputs using methods like astable multivibrators and 555 timers to evaluate performance in terms of sensitivity and stability.
3. Validate the accuracy of the developed circuits by correlating their outputs with standard LCR meter readings, ensuring the measurements are consistent and trustworthy for oil quality evaluation.
4. Investigate the impact of temperature on the dielectric constant of various oils by conducting controlled experiments, plotting results, and incorporating temperature compensation in the final analysis.
5. Propose a real-time, online method for testing cooking oil quality that is practical for daily use in kitchens, with potential for future integration into IoT-based smart monitoring systems.

Figure 3.1.2 shows a differential amplifier based Operational amplifier circuit for measuring small variations in voltage resulting from dielectric changes.

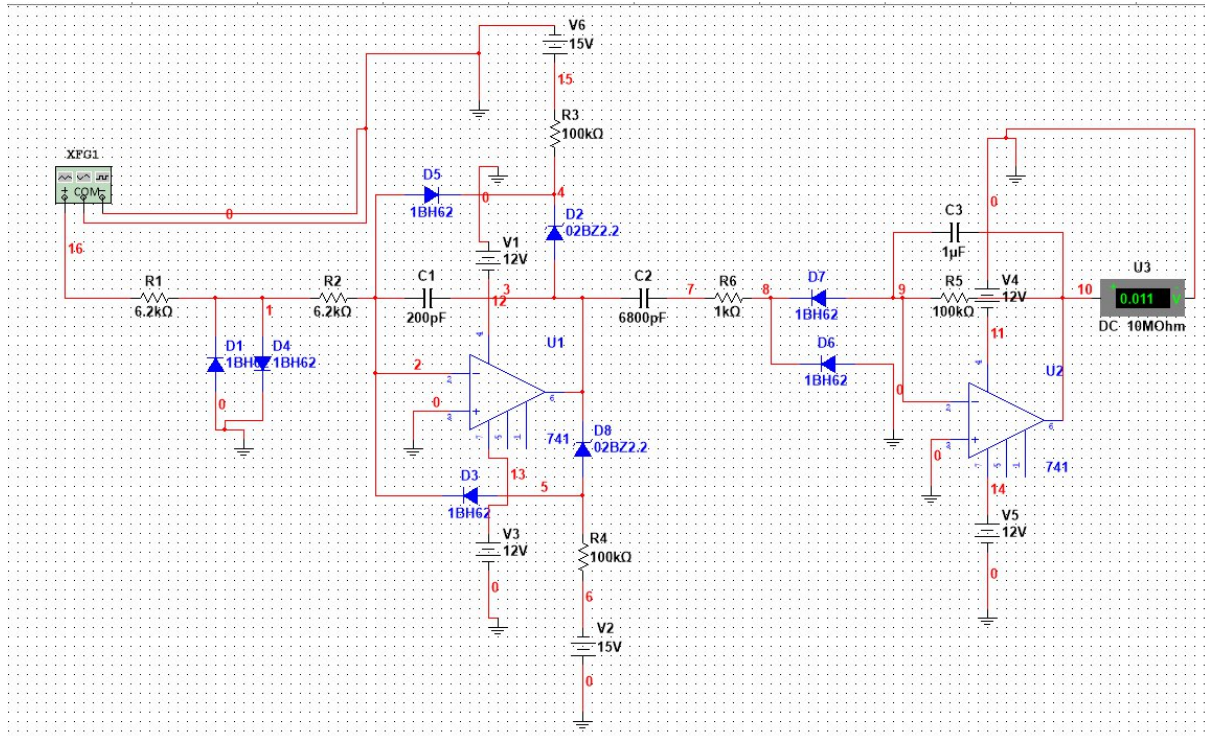


Fig 3.1.2 Differential amplifier circuit based on op-amp

Experimental procedure: Voltage readings were taken for new and old oils (coconut as well as palm oil) under different temperatures.

This technique suffered limitations regarding linearity and sensitivity, particularly because of infinitesimal changes within the picofarad order.

3.2 CAPACITANCE TO FREQUENCY (C-F) CONVERSION BASED ON 555 TIMER

To overcome the voltage sensitivity problems, a C-F approach was adopted via a 555 timer IC that was operated in Astable mode. This circuit converts capacitance variations directly to frequency changes to facilitate more accurate measurement.

Design: The 555 timer's time can be varied by varying R and C values. Since the capacitive sensor is used in place of C, variation in dielectric constant resulted in detectable changes in the frequency. The circuit for C to F converter is shown in Figure 3.2.1

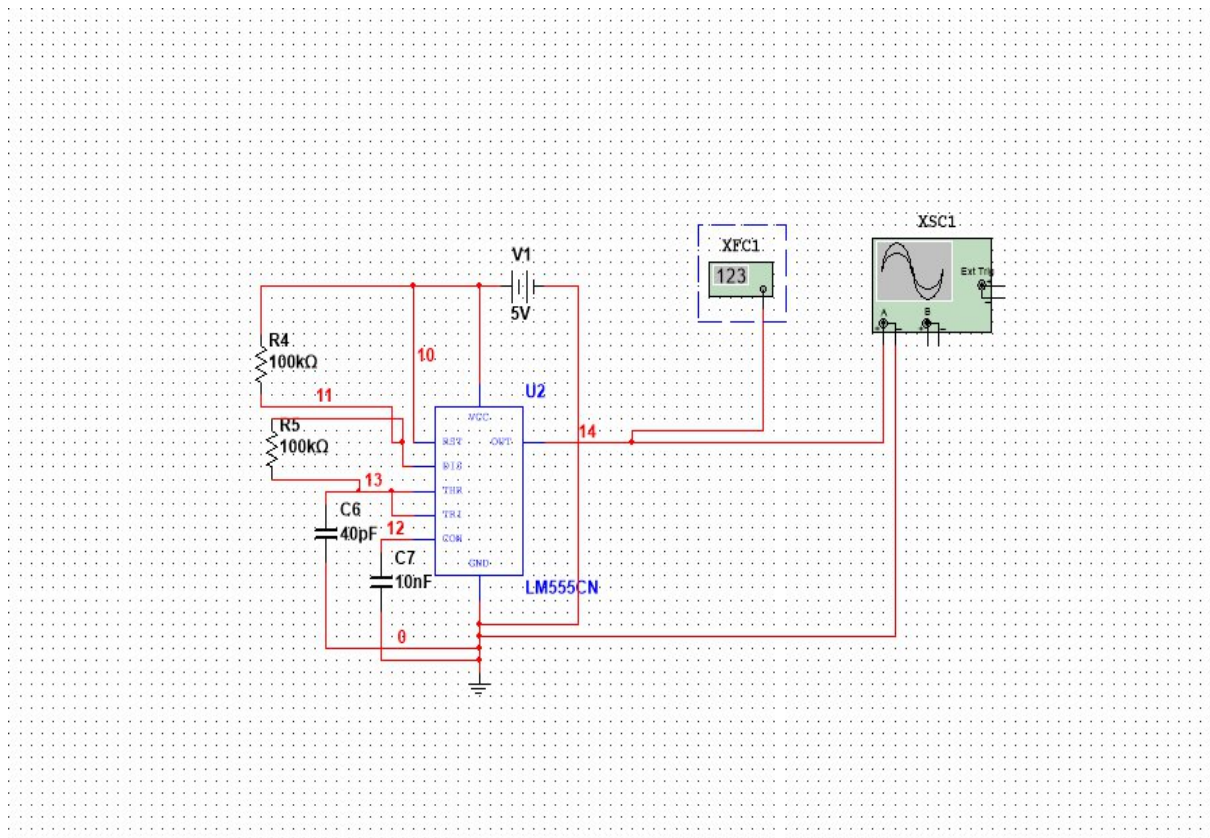


Figure 3.2.1 Capacitance to Frequency converter using 555 Timer

Hardware Setup of C-F Circuit

The C-F conversion circuit is constructed using a 555 timer IC in astable mode. The essential hardware elements are:

- 555 Timer IC (NE555)
- Resistors (chosen according to the preferred frequency range)
- Interleaved capacitive sensor
- Standard capacitor
- Breadboard for constructing the circuit
- 5V DC Regulated Power Supply
- Digital Storage Oscilloscope (DSO) for frequency output measurement

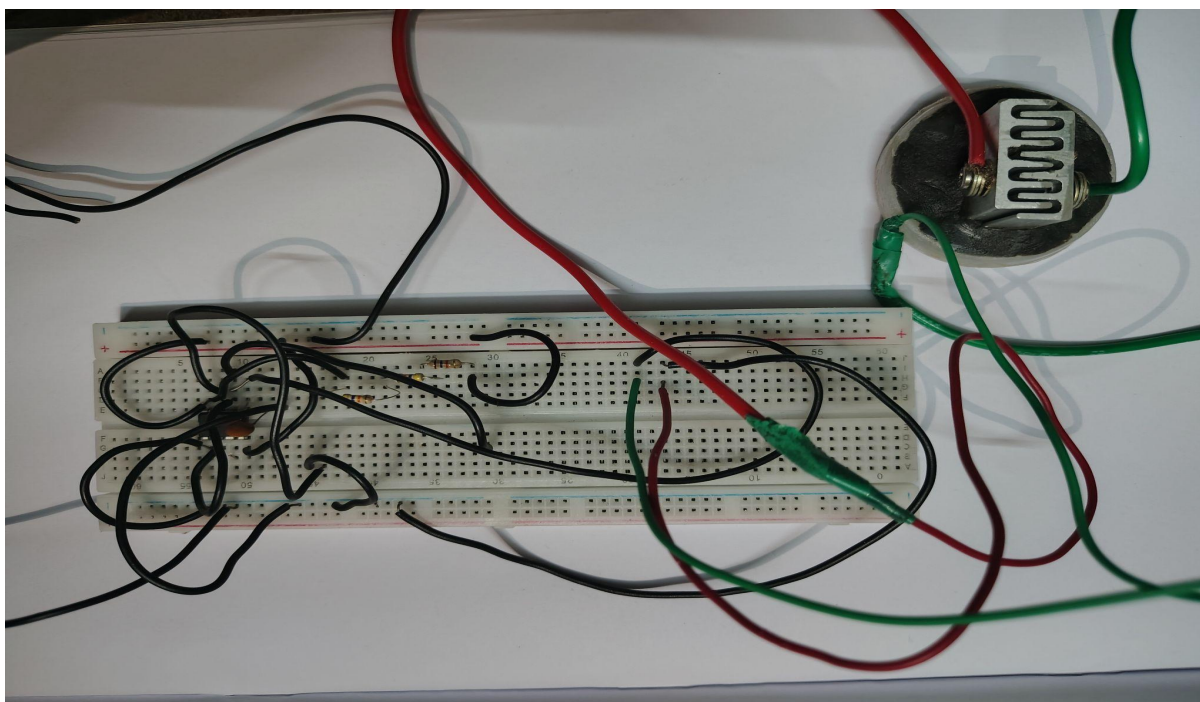


Fig 3.2.2 Capacitance to Frequency converter using 555 Timer hardware setup

The capacitive sensor served as the timing capacitor. The output frequency varied as the capacitance varied depending upon the temperature and medium. Frequency and capacitance are inversely proportional to each other, i.e. if capacitance increases frequency should decrease which allows an accurate estimation.

Some limitations have been observed in the circuit. Without oil sample there was some significant changes in frequency as capacitance varied but when tested with oil samples, there was no variations between the frequencies for different oil samples. This might be due to the less sensitive nature of the circuit. Since the changes in the capacitive sensor is very small in the order of picofarad, variations were not observed properly.

3.3 ESTIMATION OF DIELECTRIC USING SCHERING BRIDGE

The dielectric response of the oils was theoretically investigated and interpreted using the Schering bridge technique. The bridge arrangement is specifically designed to measure the dielectric loss and capacitance of insulating materials. The Figure 3.3.1 shows circuit of Schering bridge.

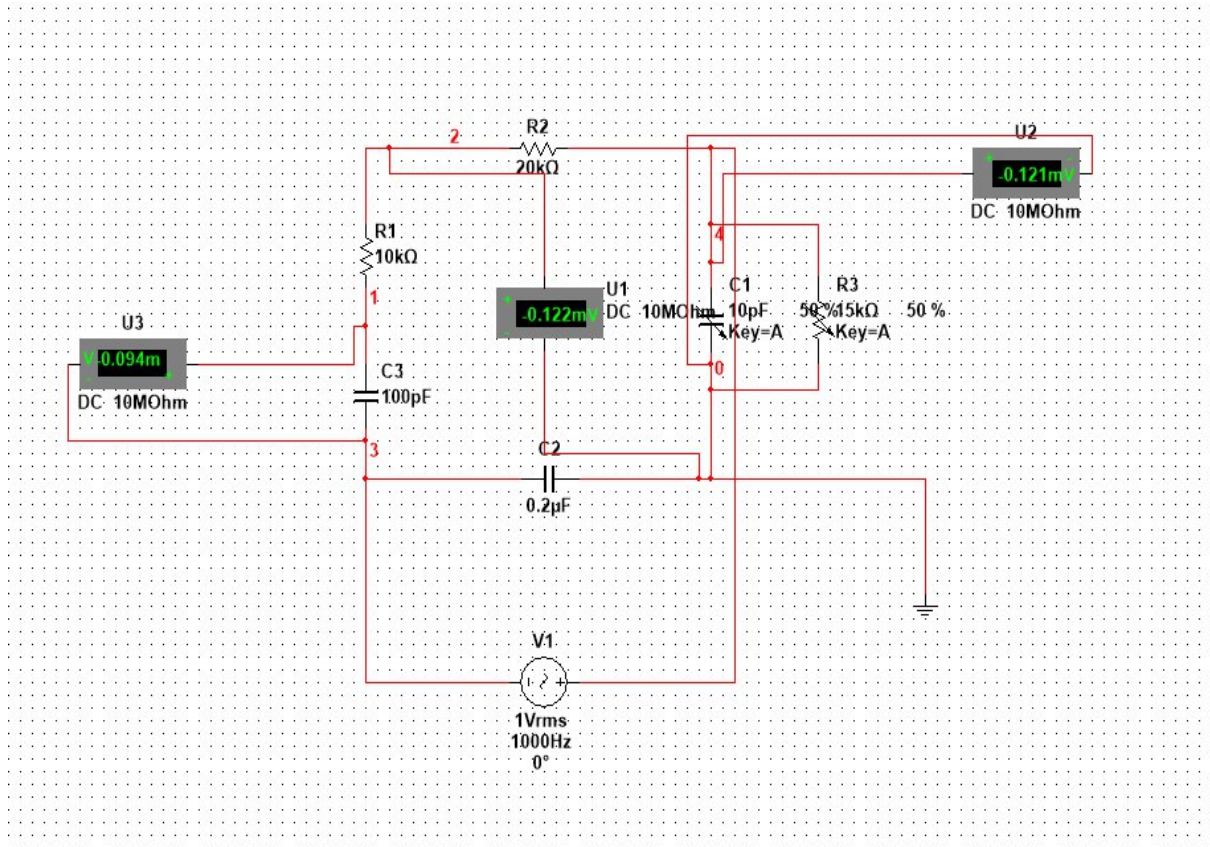


Fig 3.3.1 Schering bridge

The Schering bridge operates by balancing a known resistor and capacitor against unknown components in the oil-dipped sensor leg, allowing for the measurement of dielectric loss and capacitance. In this case, although the bridge was not physically built, the behaviour was simulated to study how changes in the dielectric loss angle and capacitance would affect the balance condition. During the simulation, it was initially planned to observe voltage variations across the capacitor to detect changes in oil properties, but unable to identify any noticeable differences. The bridge might not have been sensitive enough to detect minor variations in dielectric loss or capacitance, particularly if the oil sample's properties didn't change significantly.

3.4 DIELECTRIC CONSTANT ESTIMATION USING LCR METER

The LCR meter has an inbuilt circuit with components inductor(L), capacitor(C) and resistor(R) with which it is possible to determine the voltage across the capacitor or capacitive sensor. This meter has different modes in it to operate at different range of capacitances like at

Pico, nano and micro. Since for oils Picofarad ranges is preferred, we use Pico range. The hardware setup for LCR meter is shown in Figure 3.4.1.

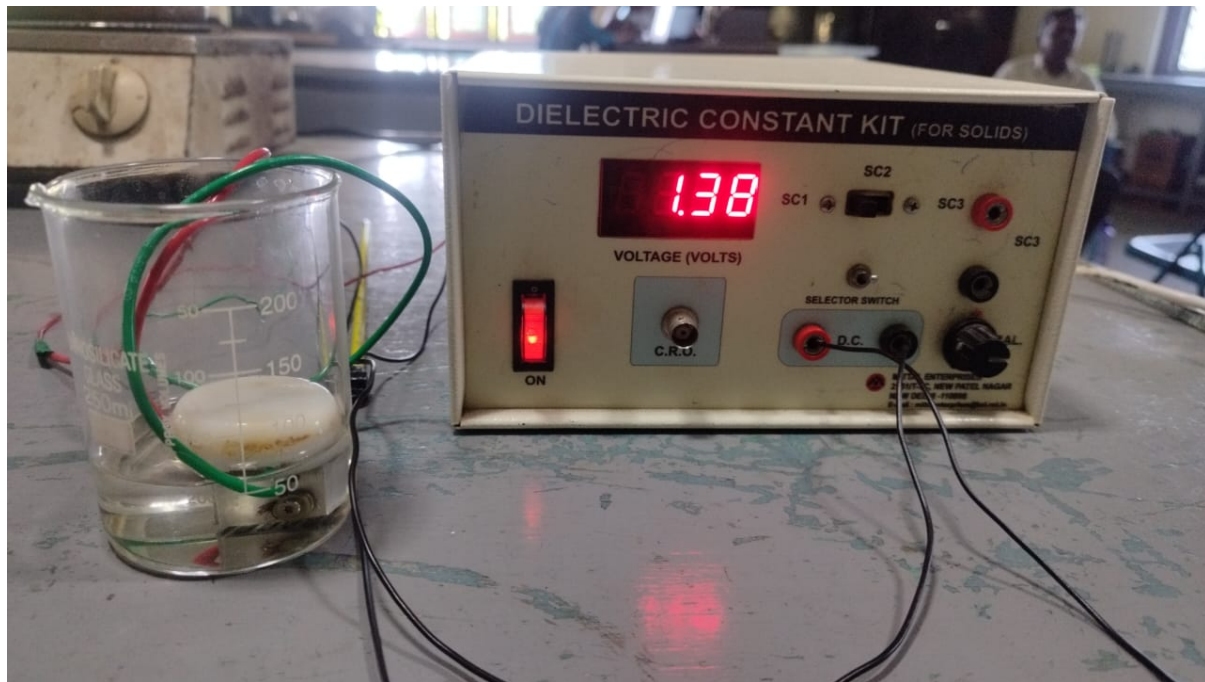


Fig 3.4.1 LCR meter setup

We have used different oil samples like coconut oil, palm oil etc. for testing and for finding the dielectric constant for those oils. To find the dielectric constant, these oil samples are heated at different temperatures say from room temperature and gradually voltage reading was taken for every 5 degrees. The voltages for different temperatures are shown below in Table 3.3.1.

Table 3.3.1: Coconut oil dielectric constant using LCR meter

Temperature (° C)	Voltage across standard capacitance (V)	Voltage across design capacitance (V)	Dielectric constant
30	1.32	1.30	4.4019
35	1.41	1.39	4.3970
40	1.41	1.41	4.3520
45	1.41	1.42	4.3047
50	1.39	1.45	4.1558
55	1.40	1.44	4.2143

60	1.39	1.44	4.1547
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Table 3.3.2: Palm oil dielectric constant using LCR meter

Temperature	Voltage across standard capacitance	Voltage across design capacitance	Dielectric constant
30	1.33	1.27	4.540
35	1.38	1.48	4.042
40	1.38	1.49	4.015
45	1.37	1.51	3.933
50	1.39	1.52	3.964
55	1.40	1.48	4.100
60	1.40	1.49	4.073

Calculation:

$$\text{Capacitance}(C) = (V_{sc}/V_{dc}) * C_{sc}$$

Where C_{sc} is capacitance of standard capacitance (45pF)

V_{sc} is voltage across standard capacitance,

V_{dc} is voltage across design capacitance

After calculating the capacitance, we can find dielectric constant (ϵ_r) by the formula

$$\epsilon_r = C / C_0$$

where C is capacitance in medium,

C_0 is capacitance in air medium (10.8pF referred from research paper)

For example, from Table 1, at 30 degrees

$V_{sc}=1.32$, $V_{dc}=1.30$, $C_{sc}=45\text{pF}$ and $C_0=10.38\text{pF}$

Therefore, $C = (1.32/1.30) * 45\text{pF}$

$$=45.692\text{pF}$$

Then $\epsilon_r = 45.693\text{pF}/10.38\text{pF}$

$$=4.401956$$

The Figures 3.4.2 and 3.4.3 shows graph of temperature vs dielectric constant of coconut oil and palm oil samples respectively.

COCONUT OIL :

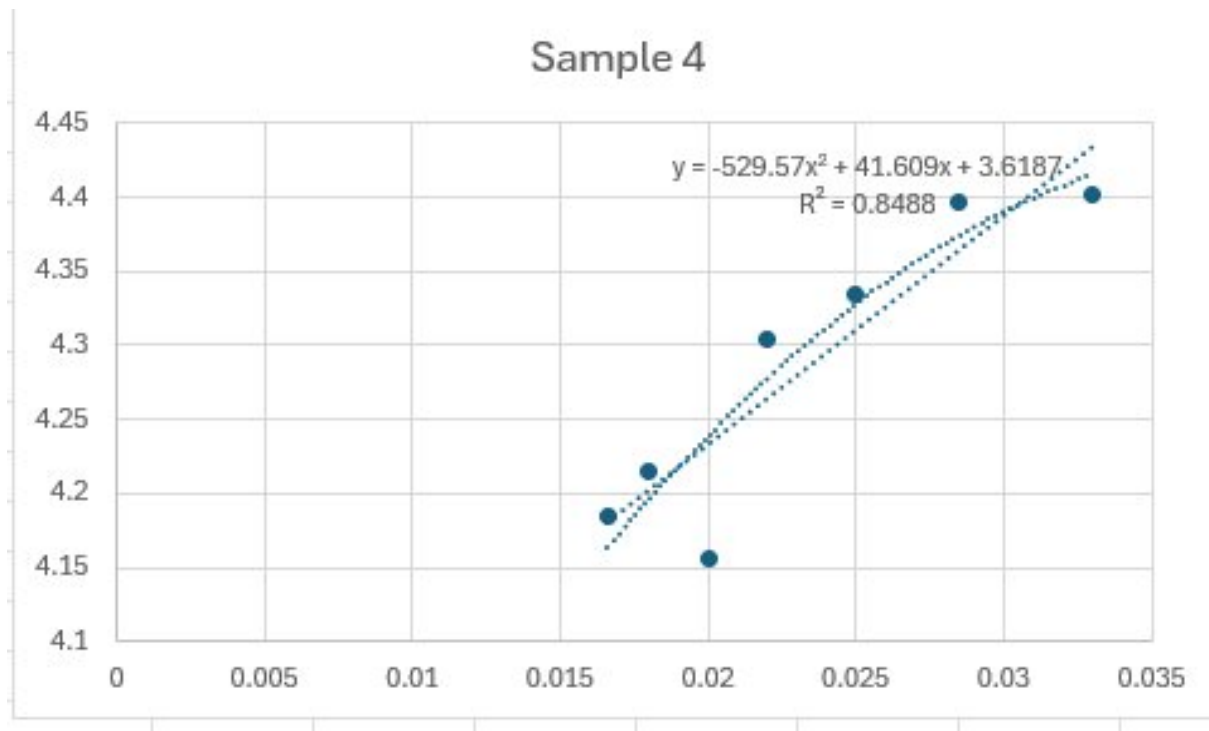


Fig 3.4.2 Graph of Temperature(1/T) vs dielectric constant of coconut oil

PALM OIL :

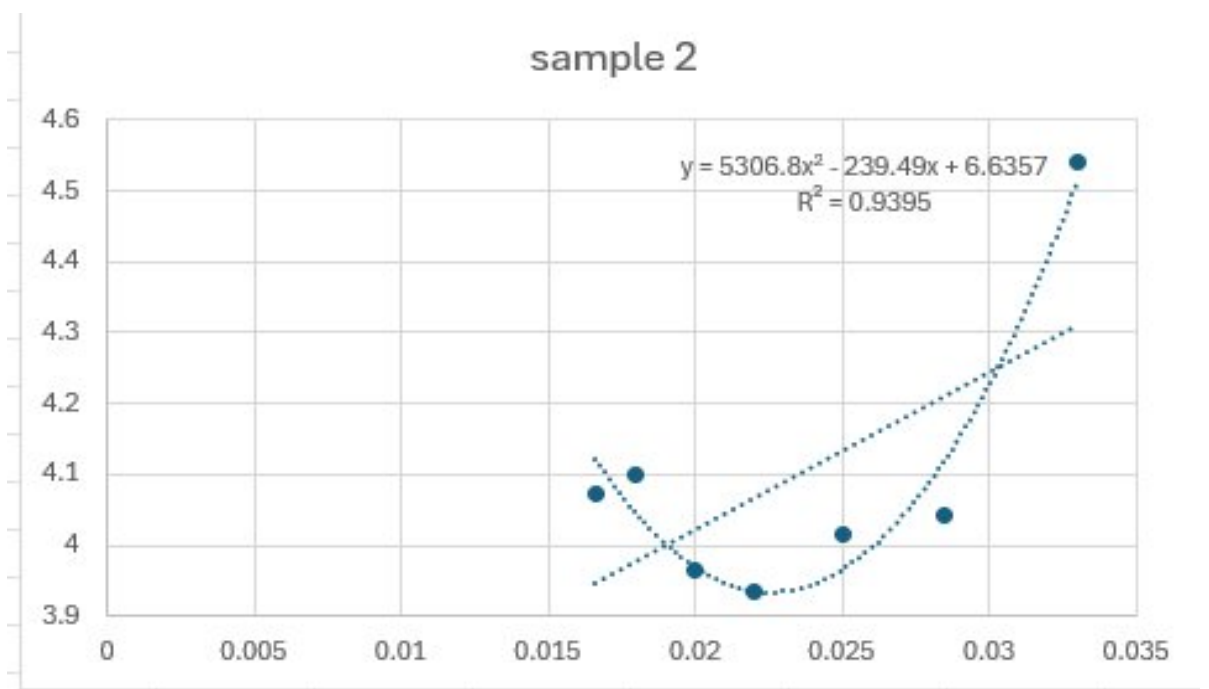


Fig 3.4.3 Graph of temperature(1/T) vs dielectric constant of palm oil

3.5 DIELECTRIC CONSTANT ESTIMATION AND WIEN BRIDGE OSCILLATOR

A third approach used was the application of a Wien Bridge Oscillator to produce stable voltage variation across capacitance for small capacitance changes. This was especially useful for fine-tuning measurements in the picofarad range. The Figure 3.5.1 shows the Wien bridge circuit.

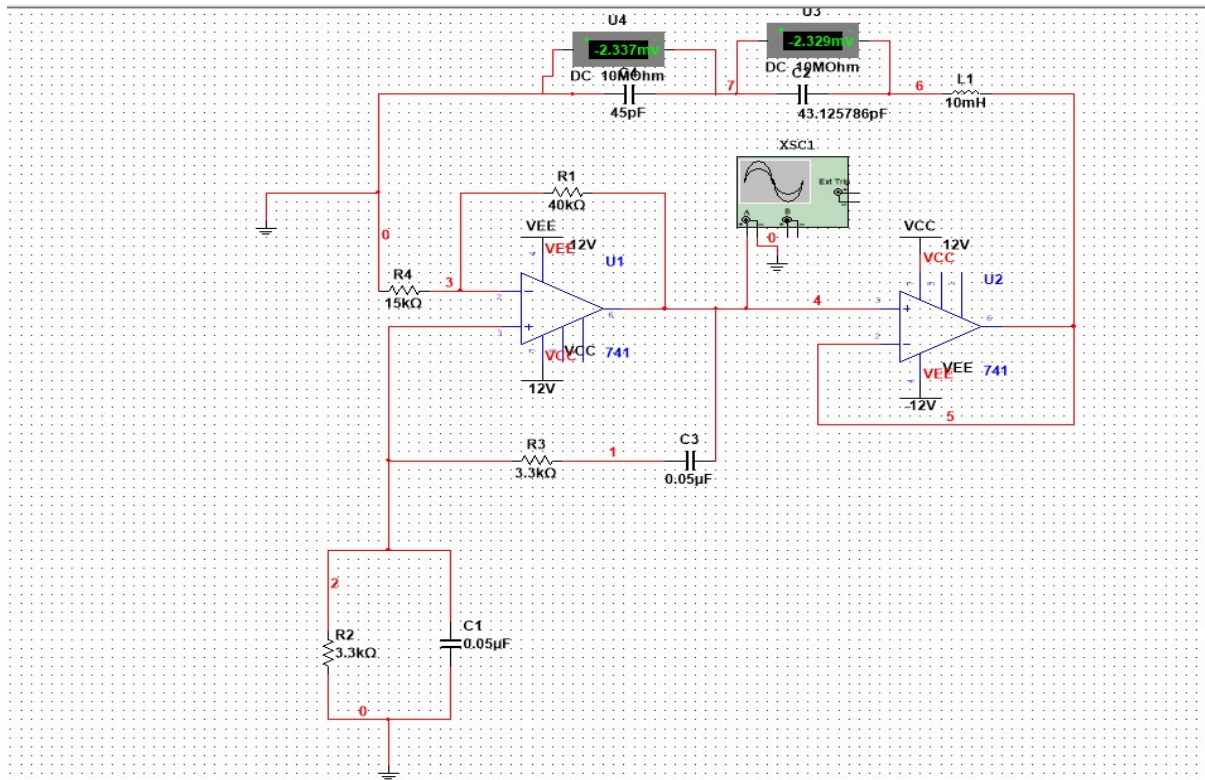


Fig 3.5.1 Capacitance to voltage using Wein bridge

The Wein bridge oscillator is designed for the frequency of 1KHz by choosing capacitance value as 0.05uF (micro farad).The formula for finding resistance value to be used in bridge is

$$f = 1 / (2 * \pi * R * C)$$

where R is resistance to be used in the bridge,

C is capacitance used in the bridge,

f is frequency of the bridge

Like LCR meter voltage across the design and standard capacitance has been measured. Here 45pF is a standard capacitance and the other is the design capacitance which is a capacitive

sensor. The design capacitance value has been calculated from the LCR meter ‘C’ and it is used in the simulation for measuring the voltage across the capacitor.

Table 3.5.1: Coconut oil dielectric constant using Wien bridge circuit

Temperature (° C)	Voltage across standard capacitance(mV)	Voltage across design capacitance(mV)	Dielectric Constant
30	2.610	2.601	4.3502
35	2.612	2.604	4.3485
40	2.613	2.613	4.3352
45	2.610	2.614	4.3286
50	2.616	2.646	4.2861
55	2.620	2.650	4.2861
60	2.588	2.645	4.2418

Table 3.5.2: Palm oil dielectric constant using Wien bridge circuit

Temperature (° C)	Voltage across standard capacitance(mV)	Voltage across design capacitance(mV)	Dielectric Constant
30	2.966	2.836	4.5330
35	2.805	2.908	4.1817
40	2.829	2.904	4.2232
45	2.825	2.908	4.2115
50	2.826	2.908	4.2130
55	2.839	2.897	4.2484
60	2.832	2.898	4.2365

CHAPTER 4

RESULTS AND DISCUSSION

4.1 GRAPH VALIDATION OF DIELECTRIC CONSTANT MEASUREMENTS (DESIGNED CIRCUIT Vs LCR METER)

Dielectric Constant Comparison for Coconut Oil: Designed Circuit vs LCR Meter

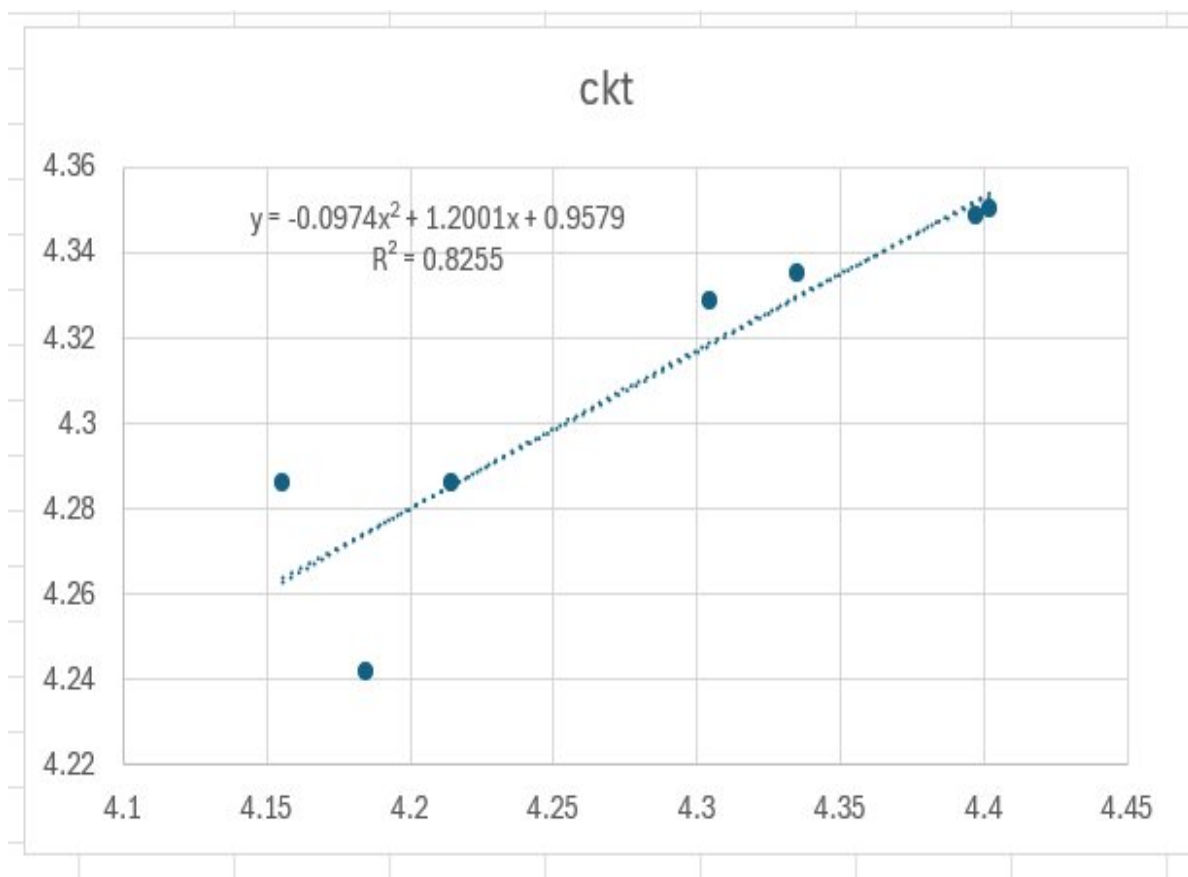


Fig 4.1.1 Graph of practical vs calculated dielectric constant of coconut oil

Figure 4.1.1 shows the graph of practical vs calculated dielectric constant of coconut oil.

Comparison of dielectric constant values measured by the developed capacitive sensing circuit and the LCR meter reference measurements, shows a strong correlation ($R^2 \approx 0.8255$).

Dielectric Constant Comparison for Palm Oil: Designed Circuit vs LCR Meter

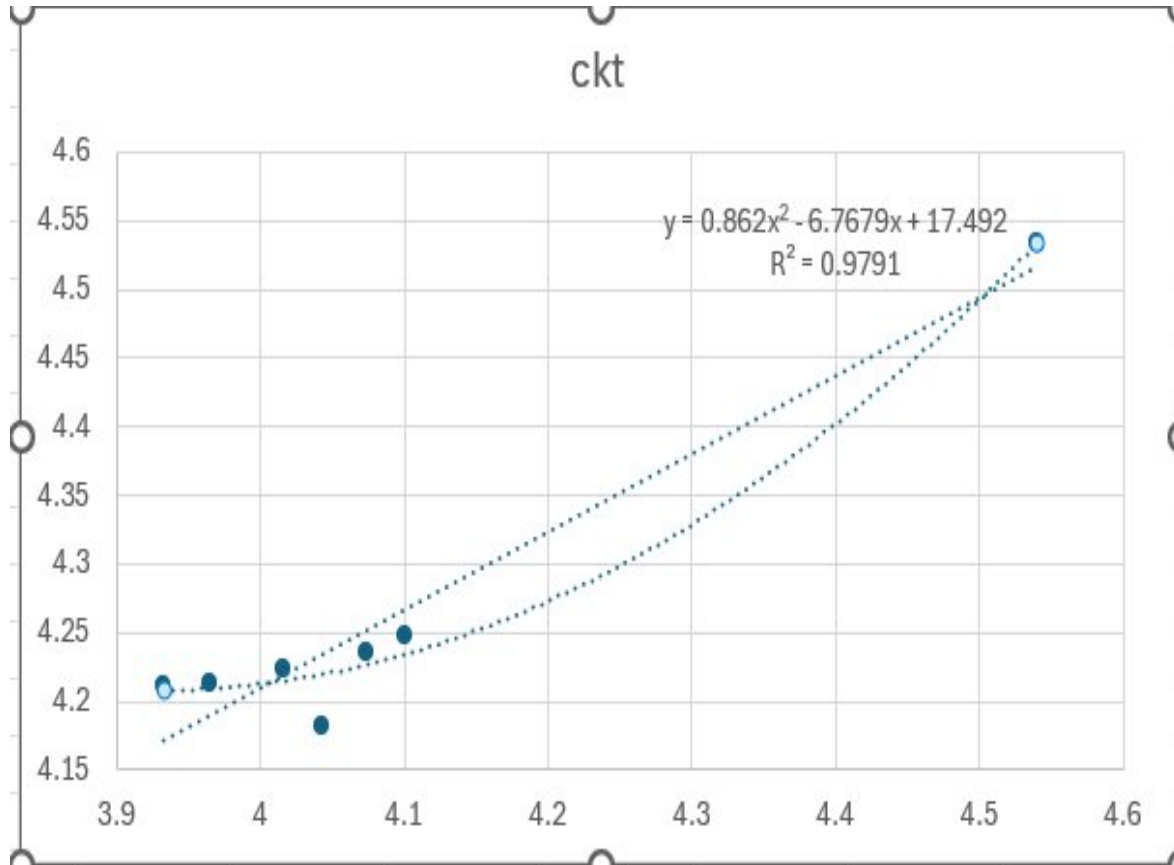


Fig 4.1.2 Graph of practical vs calculated dielectric constant of palm oil

When the dielectric constant values measured by the developed capacitive sensing circuit and the LCR meter reference measurements are compared, a strong correlation ($R^2 \approx 0.9791$) is noticed. Figure 4.1.2 displays the graph of practical vs calculated dielectric constant of palm oil.

4.2 DIELECTRIC CONSTANT VALIDATION THROUGH CIRCUIT AND STANDARD MEASUREMENT

To confirm the accuracy of the developed capacitance sensing circuit, the dielectric constant values measured by the circuit were compared to reference values obtained using a calibrated LCR meter.

With the dielectric constant values obtained by the circuit mapped on the x-axis and the LCR meter values mapped on the y-axis, the trend showed indications of a near-linear relationship, showing a strong correlation that confirmed the two sources of measurements were well intertwined.

The polynomial curve fitting provided a R^2 value just shy of 0.9, which is a good indication the circuit measurements would be considered consistent and reliable, when measured against standard laboratory equipment.

This comparison validates that the designed circuit can be effectively used for real-time, low-cost evaluation of cooking oil quality based on dielectric properties.

4.3 Frequency Output vs Capacitance (C-F Circuit)

The C-F conversion circuit based on 555 Timer indicated a linear relationship between the frequency observed and the capacitance of the oil:

Fresh oil: Outputs slightly smaller frequencies due to greater dielectric constant (greater capacitance).

Used oil: Lower dielectric constant resulted in greater output frequency.

Validation: Frequency readings were tracked and compared with theoretical expectations, affirming the accuracy of the hardware configuration.

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

1. The project investigated the various methods for the estimation of edible oils' dielectric constant and their reusability as a function of degradation.
2. Capacitance-to-frequency conversion via a 555 timer proved to be the most robust and accurate method with immunity to analogue noise.
3. The Wien Bridge Oscillator method provided good estimation at the picofarad scale and had a good correlation with theory.
4. Temperature was also an important factor in dielectric constant, where fresh oils had consistently higher values compared to the used ones.
5. Hardware circuit for frequency measurement confirmed theoretical bases and offered practical outputs conforming to predicted trends.

Future Work:

- The process can be automated by using microcontrollers (such as Arduino/ESP32) to display and record values.
- Investigate real-time wireless monitoring of dielectric values via IoT integration.
- Improve the sensitivity and miniaturization of the sensor for industrial use.
- Expand the investigation to other categories of oil and cook cycles to compile an extensive degradation database.

CHAPTER 6

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